

American College of Radiology ACR Appropriateness Criteria®

Left Lower Quadrant Pain

Variant 1: Left lower quadrant pain. Initial imaging.

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations								
							1	2	3	4	5	6	7	8	9
CT abdomen and pelvis with IV contrast	Usually appropriate	Strong	⊕⊕⊕ 1-10 mSv	⊕⊕⊕⊕ 3-10 mSv [ped]	9	9	0	0	0	0	0	0	1	3	13

References	Study Quality
23 (27271919)	3
21 (26827740)	4
24 (17765461)	2
20 (21365197)	2
2 (31208938)	3
16 (31930228)	4
19 (32172468)	3
22 (32233866)	3

CT abdomen and pelvis without IV contrast	May be appropriate	Strong	⊕⊕⊕ 1-10 mSv	⊕⊕⊕⊕ 3-10 mSv [ped]	6	6	0	0	0	0	2	10	1	3	1
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References	Study Quality
23 (27271919)	3
21 (26827740)	4
24 (17765461)	2
20 (21365197)	2
2 (31208938)	3
16 (31930228)	4

		19 (32172468)		3											
		22 (32233866)		3											
US pelvis transvaginal	May be appropriate	Limited	O 0 mSv	O 0 mSv [ped]	5	5	0	0	0	1	16	1	0	0	0
		References		Study Quality											
		18 (-3194700)		4											
US abdomen transabdominal	May be appropriate	Limited	O 0 mSv	O 0 mSv [ped]	5	5	1	0	4	3	5	4	0	0	0
		References		Study Quality											
		4 (25816854)		4											
		20 (21365197)		2											
		25 (30616263)		4											
		16 (31930228)		4											
MRI abdomen and pelvis without IV contrast	May be appropriate	Limited	O 0 mSv	O 0 mSv [ped]	4	4	0	2	6	5	4	0	0	0	0
		References		Study Quality											
		1 (25884745)		4											
		12 (23319717)		4											
MRI abdomen and pelvis without and with IV contrast	May be appropriate	Limited	O 0 mSv	O 0 mSv [ped]	4	4	0	2	6	3	4	1	1	0	0
		References		Study Quality											
		1 (25884745)		4											
		12 (23319717)		4											
Radiography abdomen and pelvis	May be appropriate	Expert Consensus	☼☼☼ 1-10 mSv	☼☼☼ 0.3-3 mSv [ped]	4	4	1	2	5	7	2	0	0	0	0
CT abdomen and pelvis without and with IV contrast	Usually not appropriate	Strong	☼☼☼☼ 10-30 mSv	☼☼☼☼☼ 10-30 mSv [ped]	3	3	2	4	3	2	4	1	1	0	0
		References		Study Quality											
		23 (27271919)		3											

		21 (26827740)	4												
		24 (17765461)	2												
		20 (21365197)	2												
		2 (31208938)	3												
		16 (31930228)	4												
		19 (32172468)	3												
		22 (32233866)	3												
Fluoroscopy contrast enema	Usually not appropriate	Expert Consensus	⊕⊕⊕ 1-10 mSv	⊕⊕⊕⊕ 3-10 mSv [ped]	2	2	7	5	5	0	0	0	0	0	0

Variant 2: Left lower quadrant pain. Suspected diverticulitis. Initial imaging.

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations								
							1	2	3	4	5	6	7	8	9
CT abdomen and pelvis with IV contrast	Usually appropriate	Strong	⊕⊕⊕ 1-10 mSv	⊕⊕⊕⊕ 3-10 mSv [ped]	9	9	0	0	0	0	0	0	0	4	13

References	Study Quality
23 (27271919)	3
37 (26070499)	3
9 (26526432)	4
33 (23592303)	2
34 (22752330)	2
36 (9609151)	3
35 (20054539)	3
30 (29204676)	3
8 (25972914)	4
2 (31208938)	3
29 (31439460)	3
38 (30268717)	3
16 (31930228)	4

31 (32118825)	4
19 (32172468)	3
39 (28837412)	3
32 (28981351)	3
11 (29075917)	3
3 (30859428)	4
5 (31206250)	4
40 (735943)	4
41 (-3195165)	4

CT abdomen and pelvis without IV contrast	May be appropriate	Strong	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	6	6	0	0	0	0	5	5	3	3	1
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References	Study Quality
23 (27271919)	3
37 (26070499)	3
9 (26526432)	4
33 (23592303)	2
34 (22752330)	2
36 (9609151)	3
35 (20054539)	3
30 (29204676)	3
8 (25972914)	4
2 (31208938)	3
29 (31439460)	3
38 (30268717)	3
16 (31930228)	4
31 (32118825)	4
19 (32172468)	3
39 (28837412)	3
32 (28981351)	3
11 (29075917)	3
3 (30859428)	4

		5 (31206250)		4											
		40 (735943)		4											
		41 (-3195165)		4											
MRI abdomen and pelvis without IV contrast	May be appropriate	Moderate	O 0 mSv	O 0 mSv [ped]	4	4	0	2	5	3	7	0	0	0	0
		References		Study Quality											
		43 (33607699)		Good											
MRI abdomen and pelvis without and with IV contrast	May be appropriate	Moderate	O 0 mSv	O 0 mSv [ped]	4	4	0	1	5	3	5	2	1	0	0
		References		Study Quality											
		43 (33607699)		Good											
US pelvis transvaginal	Usually not appropriate	Limited	O 0 mSv	O 0 mSv [ped]	3	3	1	4	4	3	3	1	0	1	0
		References		Study Quality											
		18 (-3194700)		4											
Radiography abdomen and pelvis	Usually not appropriate	Expert Consensus	☼☼☼ 1-10 mSv	☼☼☼ 0.3-3 mSv [ped]	3	3	2	2	7	5	1	0	0	0	0
US abdomen transabdominal	Usually not appropriate	Limited	O 0 mSv	O 0 mSv [ped]	3	3	1	3	9	2	3	0	0	0	0
		References		Study Quality											
		25 (30616263)		4											
CT abdomen and pelvis without and with IV contrast	Usually not appropriate	Strong	☼☼☼☼ 10-30 mSv	☼☼☼☼☼ 10-30 mSv [ped]	2	2	3	8	7	0	0	0	0	0	0
		References		Study Quality											
		23 (27271919)		3											
		37 (26070499)		3											
		9 (26526432)		4											
		33 (23592303)		2											
		34 (22752330)		2											

[illegible]

Variant 3: Left lower quadrant pain. Suspected complication(s) of diverticulitis. Initial imaging.

21 (26827740)	4
9 (26526432)	4
33 (23592303)	2
34 (22752330)	2
30 (29204676)	3
8 (25972914)	4
16 (31930228)	4
31 (32118825)	4
19 (32172468)	3
39 (28837412)	3
32 (28981351)	3
3 (30859428)	4
44 (21304297)	3
45 (15100111)	2
46 (11976860)	3

CT abdomen and pelvis without IV contrast	May be appropriate	Strong	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	5	5	0	0	0	0	13	4	1	0	0
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References	Study Quality
23 (27271919)	3
21 (26827740)	4
9 (26526432)	4
33 (23592303)	2
34 (22752330)	2
30 (29204676)	3
8 (25972914)	4
16 (31930228)	4
31 (32118825)	4
19 (32172468)	3
39 (28837412)	3
32 (28981351)	3
3 (30859428)	4

			44 (21304297)	3													
			45 (15100111)	2													
			46 (11976860)	3													
CT pelvis with bladder contrast (CT cystography)	May be appropriate	Limited	⊕⊕⊕⊕ 10-30 mSv	⊕⊕⊕⊕ 3-10 mSv [ped]	5	5	0	2	3	1	9	2	0	0	0		
		References		Study Quality													
		48 (26211219)		4													
		47 (15546583)		4													
MRI abdomen and pelvis without and with IV contrast	May be appropriate	Moderate	O O mSv	O O mSv [ped]	5	5	0	0	0	2	12	4	0	0	0		
		References		Study Quality													
		43 (33607699)		Good													
Fluoroscopy contrast enema	May be appropriate	Limited	⊕⊕⊕ 1-10 mSv	⊕⊕⊕⊕ 3-10 mSv [ped]	4	4	0	1	4	7	5	1	0	0	0		
		References		Study Quality													
		16 (31930228)		4													
		32 (28981351)		3													
Fluoroscopy cystography	May be appropriate	Limited	⊕⊕⊕ 1-10 mSv		4	4	1	2	2	4	7	1	0	0	0		
		References		Study Quality													
		48 (26211219)		4													
		47 (15546583)		4													
MRI abdomen and pelvis without IV contrast	May be appropriate	Moderate	O O mSv	O O mSv [ped]	4	4	0	4	2	5	5	0	1	0	0		
		References		Study Quality													
		43 (33607699)		Good													
US pelvis transvaginal	Usually not appropriate	Expert Consensus	O O mSv	O O mSv [ped]	3	3	1	7	7	2	0	0	0	0	0		

Radiography abdomen and pelvis	Usually not appropriate	Expert Consensus	☼☼☼ 1-10 mSv	☼☼☼ 0.3-3 mSv [ped]	3	3	1	5	8	1	2	0	0	0	0
US abdomen transabdominal	Usually not appropriate	Limited	○ 0 mSv	○ 0 mSv [ped]	3	3	1	4	5	5	2	0	0	0	0

References	Study Quality
49 (24366280)	3
34 (22752330)	2
16 (31930228)	4
3 (30859428)	4
50 (31272789)	4

CT abdomen and pelvis without and with IV contrast	Usually not appropriate	Strong	☼☼☼☼ 10-30 mSv	☼☼☼☼☼ 10-30 mSv [ped]	2	2	5	5	8	0	0	0	0	0	0
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References	Study Quality
23 (27271919)	3
21 (26827740)	4
9 (26526432)	4
33 (23592303)	2
34 (22752330)	2
30 (29204676)	3
8 (25972914)	4
16 (31930228)	4
31 (32118825)	4
19 (32172468)	3
39 (28837412)	3
32 (28981351)	3
3 (30859428)	4
44 (21304297)	3
45 (15100111)	2
46 (11976860)	3

Appendix Key

A more complete discussion of the items presented below can be found by accessing the supporting documents at the designated hyperlinks.

Appropriateness Category: The panel's recommendation for a procedure based on the assessment of the risks and benefits of performing the procedure for the specified clinical scenario.

SOE: Strength of Evidence. The assessment of the amount and quality of evidence found in the peer reviewed medical literature for an appropriateness recommendation.

- **References:** The citation number and PMID for the reference(s) associated with the recommendation.
- **Study Quality:** The assessment of the quality of an individual reference based on the number of study quality elements described in the reference.

RRL: Relative Radiation Level. A population based assessment of the amount of radiation a typical patient may be exposed to during the specified procedure.

Rating: The final rating (1-9 scale) for the procedure as determined by the panel during rating rounds.

Median: The median rating (1-9 scale) for the procedure as determined by the panel during rating rounds.

Final tabulations: A histogram showing the number of panel members who rated the procedure as noted in the column heading (ie, 1, 2, 3, etc.).

Additional supporting documents about the AC methodology and processes can be found at www.acr.org/ac.